The Astrophysical Journal, 820:22 (10pp), 2016 March 20 doi:10.3847/0004-637X/820/1/22
© 2016. The American Astronomical Society. All rights reserved.

QUIET-TIME SUPRATHERMAL (~0.1–1.5 keV) ELECTRONS IN THE SOLAR WIND

Jiawei Tao[1], Linghua Wang[1], Qiugang Zong[1], Gang Li[2], Chadi S. Salem[3], Robert F. Wimmer-Schweingruber[4], Jiansen He[1], Chuanyi Tu[1], and Stuart D. Bale[3,5]
[1] School of Earth and Space Science, Peking University, Beijing 100871, China; wanglhwang@gmail.com
[2] Department of Physics and CSPAR, University of Alabama in Huntsville, Alabama 35899, USA
[3] Space Sciences Laboratory, University of California, Berkeley, CA 94720, USA
[4] Institute for Experimental and Applied Physics, University of Kiel, Leibnizstrasse 11, D-24118 Kiel, Germany
[5] Physics Department, University of California, Berkeley, CA 94720, USA
Received 2015 October 21; accepted 2016 January 27; published 2016 March 11

ABSTRACT

We present a statistical survey of the energy spectrum of solar wind suprathermal (~0.1–1.5 keV) electrons measured by the *WIND* 3DP instrument at 1 AU during quiet times at the minimum and maximum of solar cycles 23 and 24. After separating (beaming) strahl electrons from (isotropic) halo electrons according to their different behaviors in the angular distribution, we fit the observed energy spectrum of both strahl and halo electrons at ~0.1–1.5 keV to a Kappa distribution function with an index κ and effective temperature $T_{\rm eff}$. We also calculate the number density n and average energy $E_{\rm avg}$ of strahl and halo electrons by integrating the electron measurements between ~0.1 and 1.5 keV. We find a strong positive correlation between κ and $T_{\rm eff}$ for both strahl and halo electrons, and a strong positive correlation between the strahl n and halo n, likely reflecting the nature of the generation of these suprathermal electrons. In both solar cycles, κ is larger at solar minimum than at solar maximum for both strahl and halo electrons. The halo κ is generally smaller than the strahl κ (except during the solar minimum of cycle 23). The strahl n is larger at solar maximum, but the halo n shows no difference between solar minimum and maximum. Both the strahl n and halo n have no clear association with the solar wind core population, but the density ratio between the strahl and halo roughly anti-correlates (correlates) with the solar wind density (velocity).

Key words: acceleration of particles – solar wind – Sun: particle emission

1. INTRODUCTION

The solar wind electrons observed near 1 au consist of three components: a thermal (~10 eV) Maxwellian core that contains ~90%–95% of the plasma density, a much hotter (~50–80 eV) halo/strahl that contains ~5%–10% of the plasma density (e.g., Montgomery et al. 1968; Vasyliunas 1968; Feldman et al. 1975; Rosenbauer et al. 1977; Pilipp et al. 1987; Maksimovic et al. 2005; Zouganelis 2008), and a power-law ($J \propto E^{-\beta}$ with $\beta \sim 2.4$) superhalo at energies above ~2 keV (Lin 1998; Wang et al. 2012; Wang & Yang 2015; Yang et al. 2015). The halo population is observed to have a Maxwellian/Kappa spectrum and isotropic angular distribution. The strahl population, predominantly observed in fast solar wind, is antisunward beaming along the interplanetary magnetic field (IMF), but its spectrum (or velocity distribution) has not been studied yet.

The generation of strahl and halo electron populations is still not well known. The highly anisotropic strahl could result from the escape of thermal electrons from the hot (~10^6 K) solar corona (e.g., Feldman et al. 1975; Salem et al. 2007) that carries heat flux outward, while the isotropic halo may be due to the scattering of the strahl in the interplanetary medium (IPM; e.g., Montgomery et al. 1968; Feldman et al. 1975; Rosenbauer et al. 1977; Pilipp et al. 1987; Pierrard et al. 2001). Combining the measurements from the *Helios*, *WIND*, and *Ulysses* spacecraft, Maksimovic et al. (2005) and Stverak et al. (2009) reported that the relative halo (strahl) number density to the total electron density increases (decreases) with the heliocentric distance, while the relative core density remains roughly constant, supporting the above scenarios. However, the strahl angular width appears to increase with the heliocentric distance in fast solar wind (Hammond et al. 1996), despite the magnetic focusing effect.

Various theories/models have been developed to explain the generation of halo/strahl electrons in the solar wind (e.g., Scudder & Olbert 1979; Scudder 1992; Maksimovic et al. 1997a; Pierrard et al. 1999; Zouganelis et al. 2004; Vocks et al. 2005; Yoon et al. 2006; Che & Goldstein 2014). Scudder & Olbert (1979) first suggested that the solar wind expansion combined with reduced Coulomb collisions for suprathermal electrons can give rise to nonthermal electron populations. Following Scudder's work, Maksimovic et al. (1997a) developed a kinetic model of solar wind that is based on Kappa distributions for electrons and protons escaping from the corona, and this model is supported by the observed anti-correlation between the solar wind speed and the fitted κ index of halo electrons (Maksimovic et al. 1997b). Vocks et al. (2005) suggested that the resonant interactions between antisunward-travelling suprathermal electrons and sunward-propagating whistler waves are capable of significantly pitch-angle (PA) scattering suprathermal electrons, leading to the formation of the halo/strahl as observed in the IPM. Yoon et al. (2006) and Kim et al. (2015) suggested that the dynamic equilibrium between suprathermal electrons and Langmuir waves/whistler waves could produce the Kappa distribution of the halo population.

In this paper, we survey the energy spectrum of strahl electrons, as well as halo electrons, measured at ~0.1–1.5 keV by the *WIND* 3D Plasma & Energetic Particle (3DP) instrument at quiet times during solar minimum and maximum of cycles 23 and 24. After fitting a Kappa distribution to both the strahl and halo spectra, we find a positive correlation between the fitted index κ and effective temperature $T_{\rm eff}$. We also find a

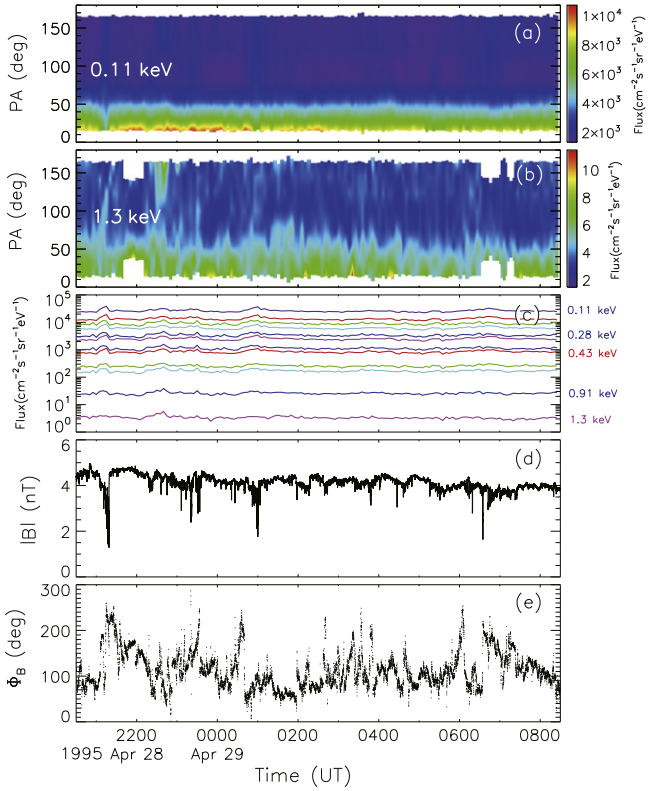


Figure 1. Example of the quiet-time ~0.1–1.5 keV electrons observed by the *WIND* 3DP instrument on 1995 April 28–29. (a) and (b): the spectrograms of electron PA distribution at 0.1 keV observed by EESA-L and at 1.3 keV by EESA-H. (c): the five-minute average flux vs. time of halo electrons after subtracting the beaming strahl electrons. The labeled energy is the center energy of the energy channel. (d) and (e): the IMF magnitude and azimuthal angle measured by the *WIND* MFI instrument.

positive correlation between the integrated number density of strahl and halo electrons at 0.1–1.5 keV.

2. DATA ANALYSIS

The *WIND* spacecraft was launched on 1994 November 1, with highly elliptical earth orbits (apogees ~60–100 R_E) or halo orbits around the Lagrange 1 point. The onboard 3DP instrument provides the full three-dimensional measurements of electrons from solar wind thermal plasma to ~400 keV (Lin et al. 1995). Silicon semiconductor telescopes (SST) measure ~20–400 keV electrons with energy channel resolution of $\Delta E/E = 0.3$, while electron electrostatic analyzers (EESA-L and EESA-H) measure ~3 eV–30 keV electrons with energy resolution of $\Delta E/E = 0.2$. The three-dimensional electron data from 3DP are binned into eight PA bins with a 22°.5 angular resolution (Wang 2009), according to the IMF direction measured by the *WIND* MFI instrument (Lepping et al. 1995). In this study, we use the EESA (EESA-L and EESA-H) electron measurements to survey the suprathermal electrons in the solar wind during quiet times at the minimum (1995 April to 1997 August, 2006 June to 2010 September) and maximum (1999 June to 2000 July, 2011 June to 2014 January) of solar cycles 23 and 24, after correction for the spacecraft potential (Salem et al. 2001; although the spacecraft potential has a very small effect on the electron measurements at 0.1–1.5 keV).

As defined by Wang et al. (2015), a quiet time is a >12 hr time period when the electron measurements at ~20–200 keV (by SST) show no significant temporal variation. Here we investigate the quiet-time observations at ~0.1–1.5 keV by EESA, that are dominated by strahl/halo electrons in the solar wind. Figure 1 shows an example of quiet-time EESA electron

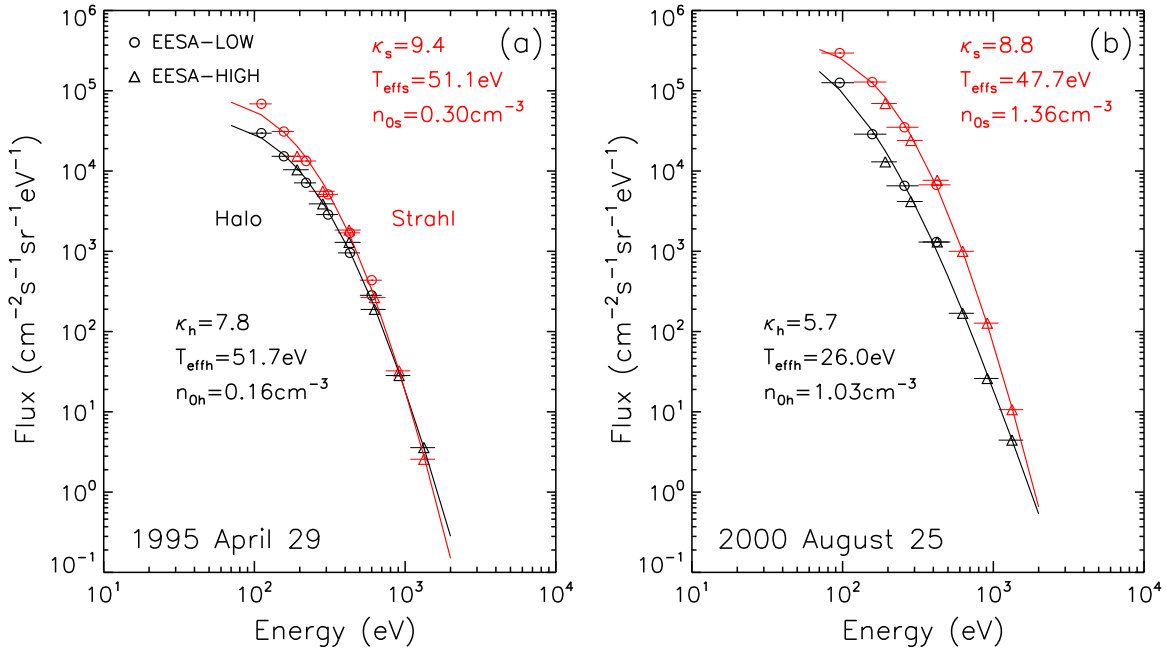


Figure 2. Examples of the energy spectrum of the 0.1–1.5 keV strahl (red) and halo (black) electrons observed by EESA-L (circles) and -H (triangles) on 1995 April 29 (left) and 2000 August 25 (right).

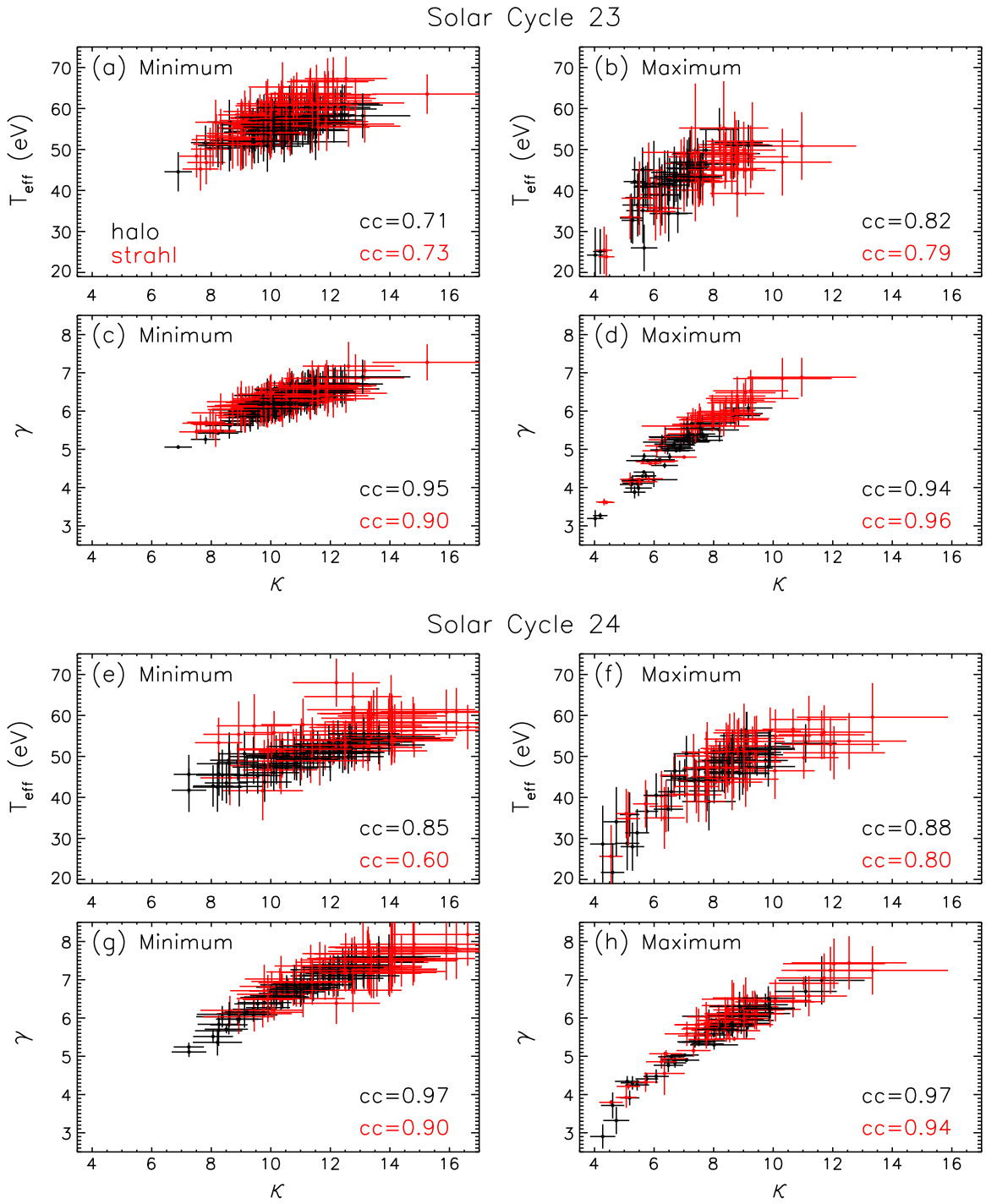


Figure 3. Scatter diagrams of $T_{\rm eff}$ vs. κ and γ vs. κ, for the quiet-time strahl (red) and halo (black) electrons observed at solar minimum (left) and maximum (right) in cycles 23 and 24. The error bars represent the estimated fitting uncertainties. The γ index is the fitted power-law spectral index of electrons at ~0.5–1.5 keV.

measurements at ~0.1–1.5 keV on 1995 April 28–29. As the IMF points antisunward (panel (e)), the strahl electrons are beaming along the direction of IMF with PA $\lesssim 50°$, while the isotropic halo electrons appear to dominate the electron measurements at other PAs (see panel (a) and (b)).

We can separate (beaming) strahl electrons from (isotropic) halo electrons based on their different anisotropies shown in the angular distribution. In the selected quiet-time samples, the strahl electrons generally show a PA width $\lesssim 45°$. When the observed IMF points antisunward (sunward), we average the electron data in the six PA bins from 45° to 180° PA (from 0° to 135° PA) to obtain the flux of halo electrons, J_h, and subtract J_h from the average flux in the other two PA bins from 0° to 45° PA (from 135° to 180° PA) to obtain the flux of strahl electrons, J_s. Figure 1(c) shows the flux versus time of halo electrons measured from 0.11 keV to 1.3 keV (channel center energy) on 1995 April 28–29.

We identify the quiet-time observations with the observed $J_s \geqslant 0.5 \times J_h$ at all energies from ~0.1 to 1.5 keV. We find 89 (58) quiet-time days at solar minimum and 46 (52) quiet-time days at solar maximum in cycle 23 (24). For each quiet-time

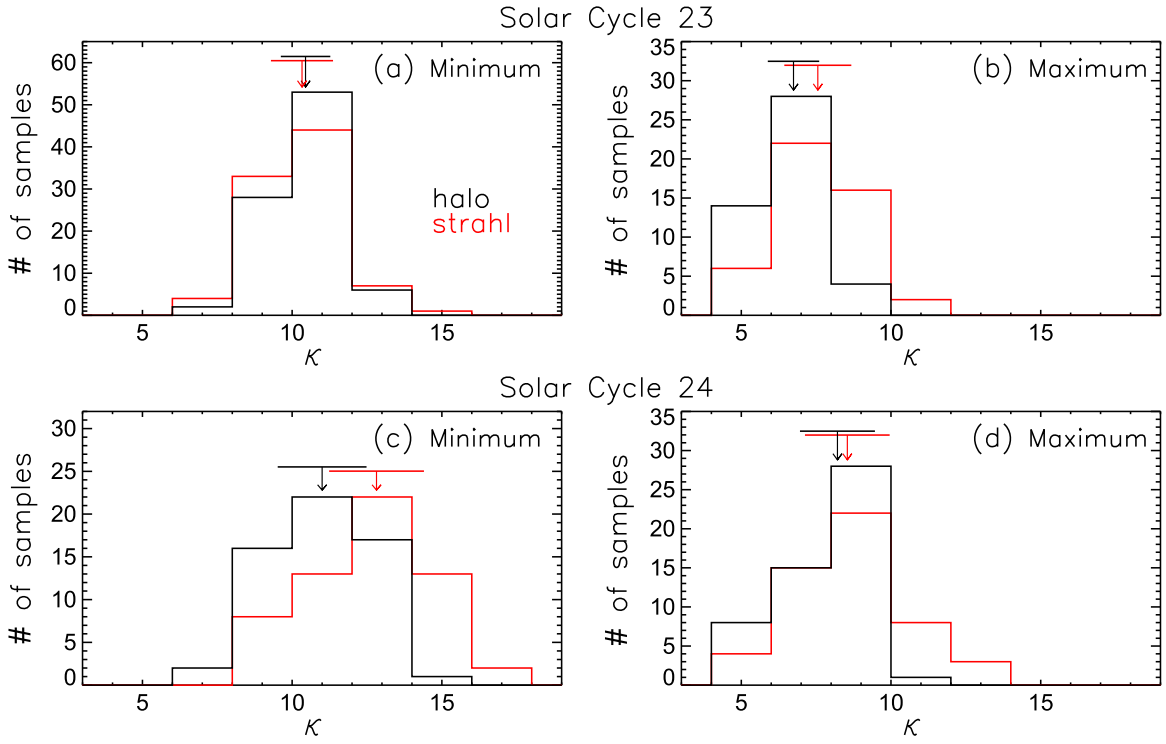


Figure 4. Histograms of the fitted κ for the 0.1–1.5 keV strahl (red) and halo (black) electrons observed at solar minimum (left) and maximum (right) in cycles 23 (top) and 24 (bottom). The arrows indicate the median, while the horizontal lines (at the start of arrow) show the corresponding median absolute deviation.

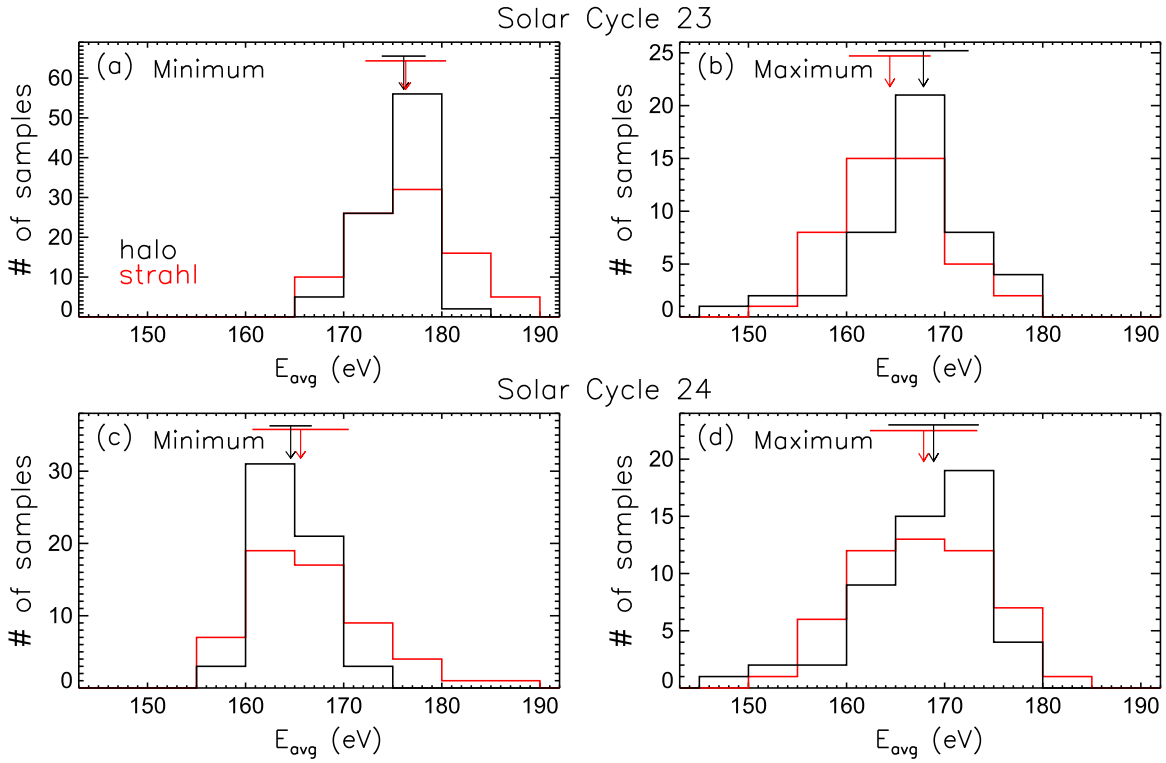


Figure 5. Histograms of the average energy $E_{\rm avg}$ for the 0.1–1.5 keV strahl (red) and halo (black) electrons observed at solar minimum (left) and maximum (right) in cycles 23 (top) and 24 (bottom). The arrows indicate the median, while the horizontal lines (at the start of the arrow) show the corresponding median absolute deviation.

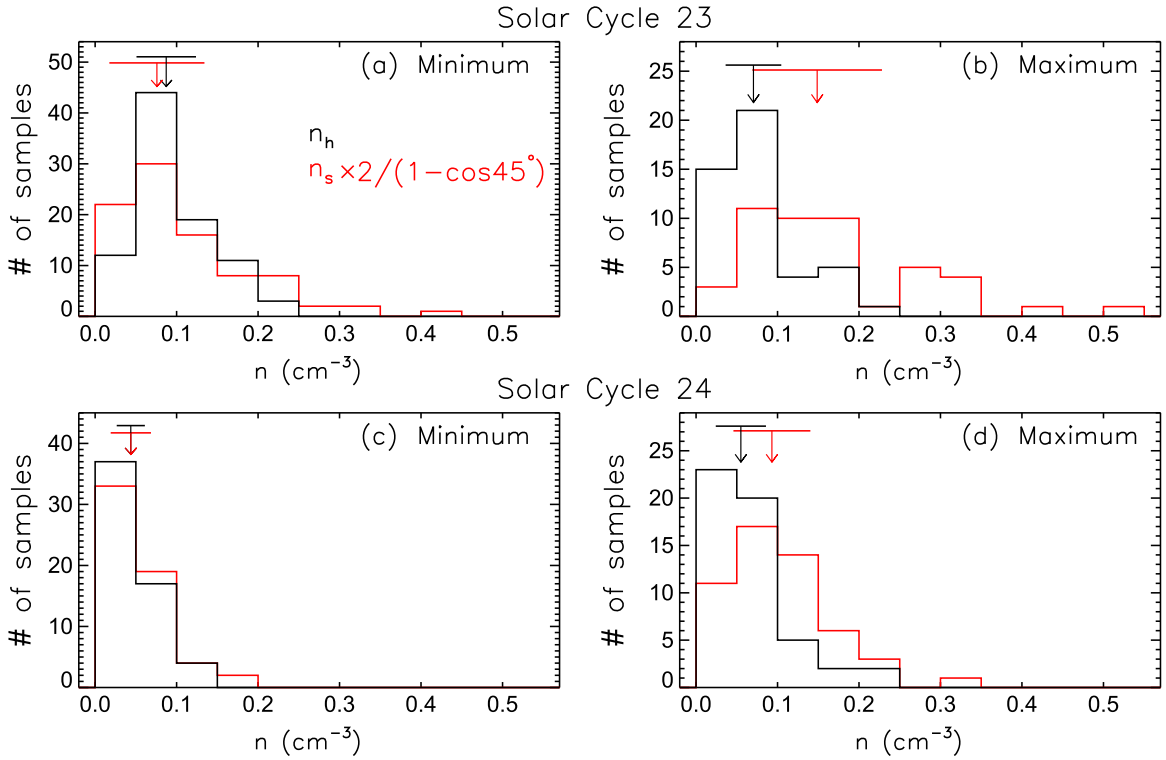


Figure 6. Histograms of the number density n for the 0.1–1.5 keV strahl (red) and halo (black) electrons observed at solar minimum (left) and maximum (right) in cycles 23 (top) and 24 (bottom). The shown strahl n is multiplied by a factor of $2/(1-\cos 45^\circ)$. The arrows indicate the median, while the horizontal lines (at the start of the arrow) show the corresponding median absolute deviation.

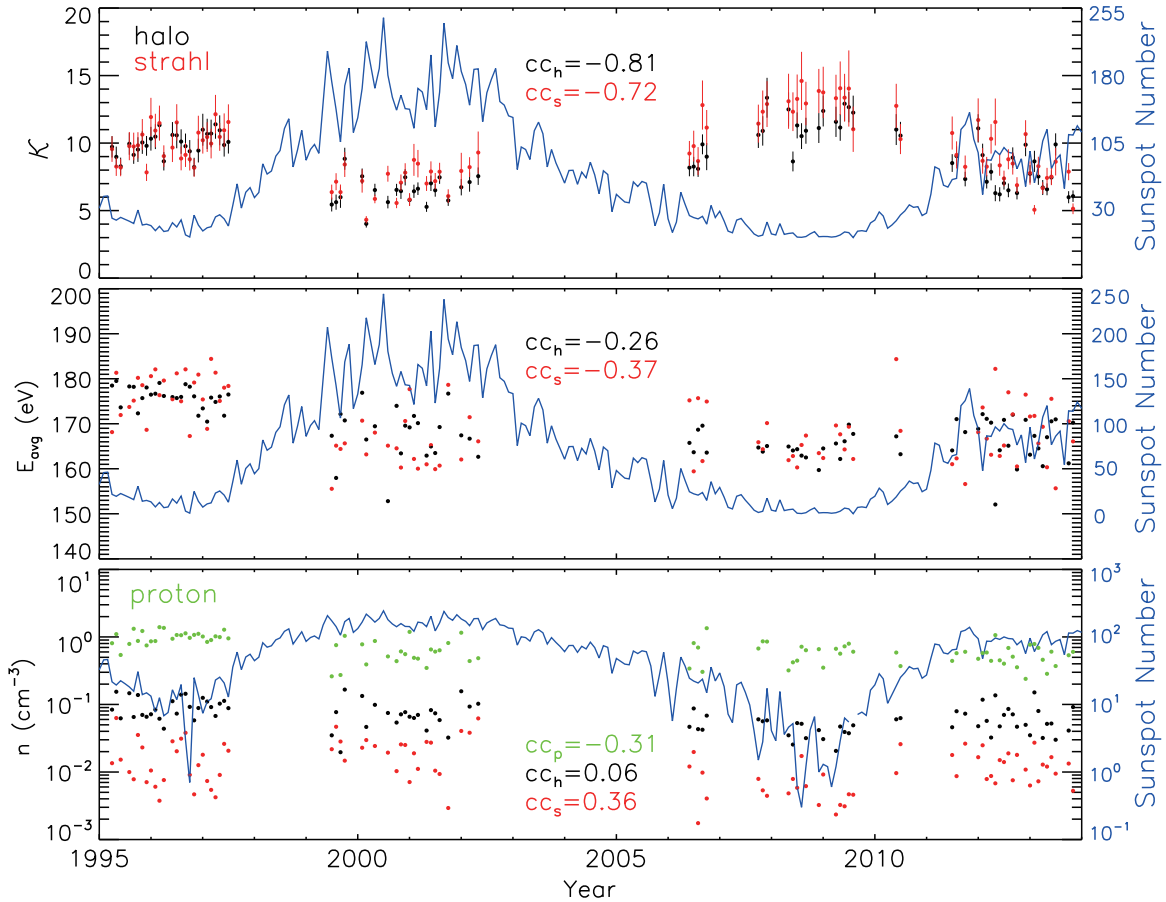


Figure 7. Temporal variation of κ (a), E_{avg} (b), and n (c) for the monthly average strahl (red) and halo (black) electrons at 0.1–1.5 keV observed during quiet times. The blue curves show the monthly sunspot number. The estimated uncertainties of κ, E_{avg}, and n are plotted.

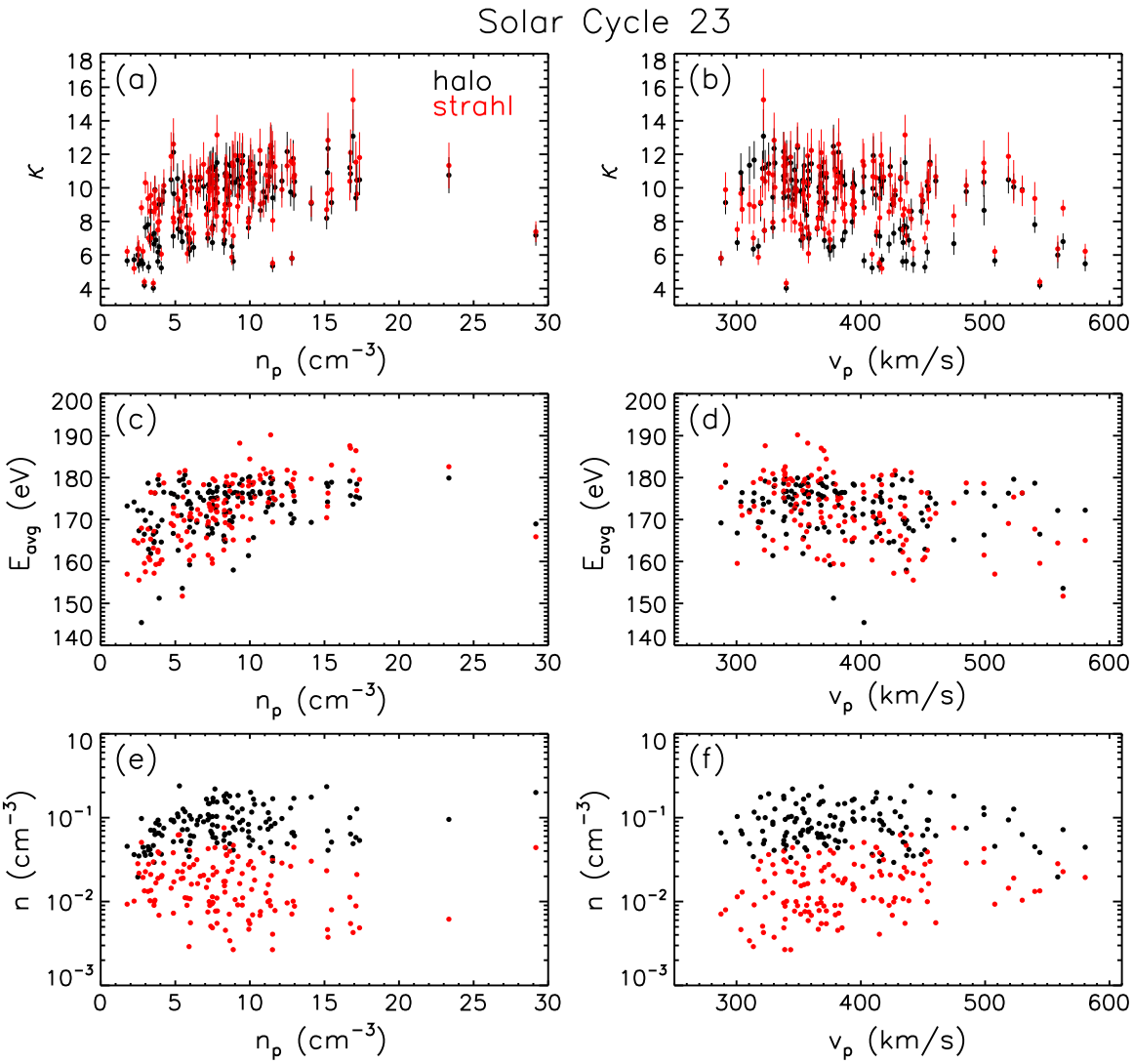


Figure 8. (a) and (b): scatter diagrams of the strahl (red) and halo (black) κ vs. the solar wind proton density n_p and velocity v_p, for solar cycle 23. (c) and (d): same scatter diagrams, but for the $E_{\rm avg}$. (e) and (f): Same scatter diagrams, but for the n. The estimated uncertainties of κ, $E_{\rm avg}$, and n are plotted.

day, we select a 12-hr sample to average the halo/strahl electron observations at ~0.1–1.5 keV from EESA, and fit the average electron fluxes to a Kappa distribution described as

$$J(E) = \frac{En_0}{\pi m_e^{1/2}((2\kappa-3)k_{\rm B}T_{\rm eff})^{3/2}}\frac{\Gamma(\kappa+1)}{\Gamma(\kappa-\frac{1}{2})} \times\left(1+\frac{2E}{(2\kappa-3)k_{\rm B}T_{\rm eff}}\right)^{-(\kappa+1)}, \quad (1)$$

where m_e is the electron mass, κ (>3/2) is the Kappa index, n_0 is the number density of the entire distribution, and $T_{\rm eff}$ is the kinetic temperature defined in the Tsallis statistical mechanics for stationary states (Livadiotis & McComas 2010). When κ increases toward infinity, the Kappa function becomes a thermal Maxwellian distribution, while when κ decreases toward 3/2, its high-energy tail becomes a power law.

Figure 2(a) plots the flux versus energy of strahl (red) and halo (black) electrons observed at ~0.1–1.5 keV by EESA-L (circles) and EESA-H (triangles) on 1995 April 29, near the solar minimum of cycle 23. Compared to the halo electrons, the flux of strahl electrons is significantly larger at energies below ~0.7 keV, but is similar at energies above. The fitted κ is 9.37 ± 0.98 (7.81 ± 0.49) and $T_{\rm eff}$ is 51.1 ± 5.9 eV (51.7 ± 5.7 eV) for the strahl (halo). Figure 2(b) shows the quiet-time strahl and halo electron spectra measured on 2000 August 25, near the solar maximum of cycle 23. In this quiet-time sample, the strahl electron flux is larger than the halo flux at all energies from ~0.1 to 1.5 keV. The fitted κ is 8.82 ± 0.59 (5.67 ± 0.45) and $T_{\rm eff}$ is 47.7 ± 3.9 eV (26.0 ± 5.7 eV) for the strahl (halo). Note that in both samples, the fitted halo κ is clearly smaller than the strahl κ.

Figure 3 shows a clear positive correlation between the fitted index κ and temperature $T_{\rm eff}$ for both strahl (red) and halo (black) electrons at quiet times. Such a positive correlation may provide stringent constraints on the nature of generation processes of strahl and halo electrons in the solar wind. We also fit the electron differential fluxes at ~0.5–1.5 keV to a power-law function, $J \sim E^{-\gamma}$. Figure 3 shows a strong positive correlation between the γ and κ for both the strahl and halo, suggesting that a smaller (larger) κ corresponds to a more (less) efficient electron acceleration (at least at ~0.5–1.5 keV). In solar cycle 23, the strahl (halo) κ index ranges from 7.5 to 15.3

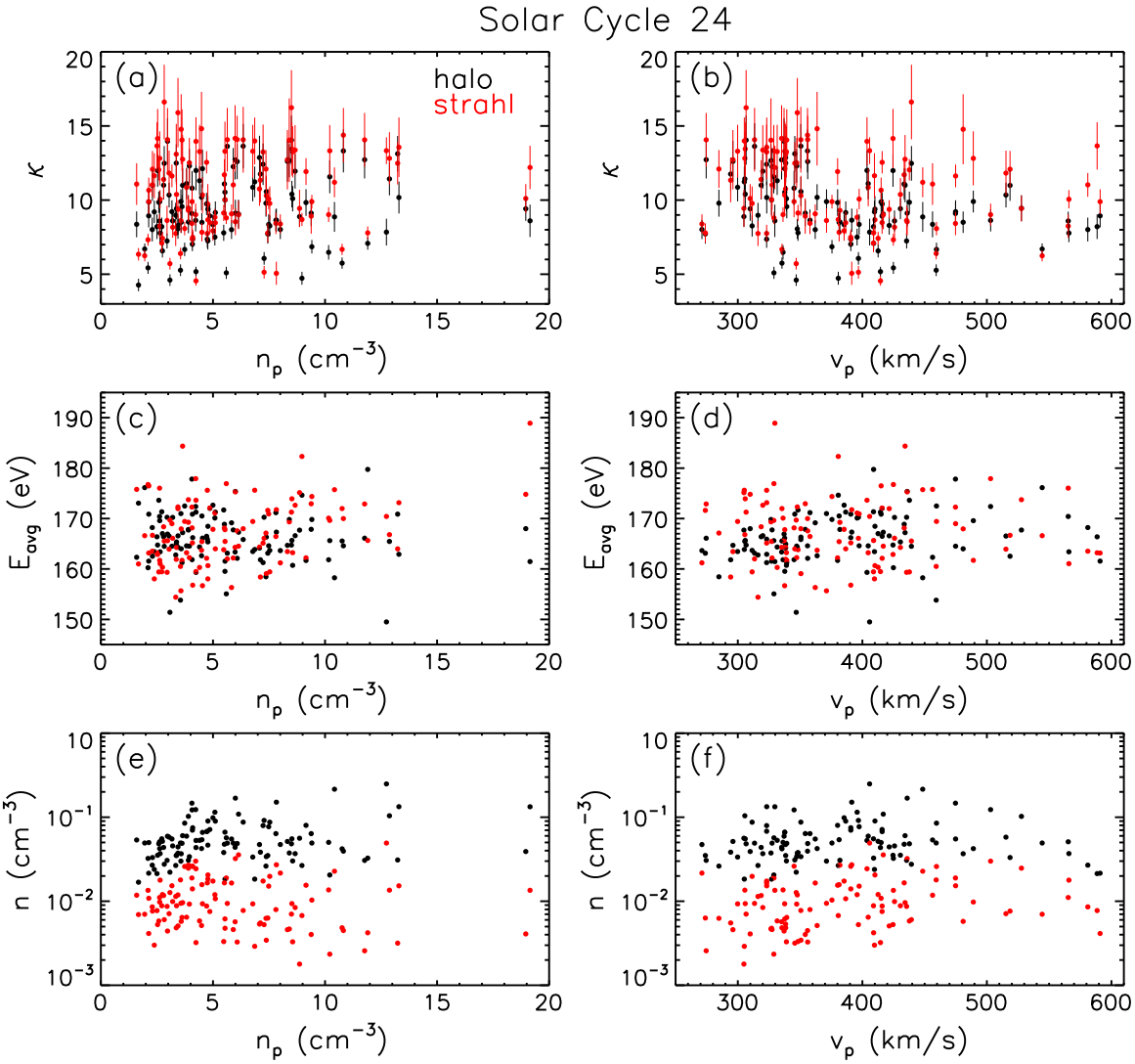


Figure 9. (a) and (b): scatter diagrams of the strahl (red) and halo (black) κ vs. the solar wind proton density n_p and velocity v_p, for solar cycle 24. (c) and (d): same scatter diagrams, but for the $E_{\rm avg}$. (e) and (f): same scatter diagrams, but for the n. The estimated uncertainties of κ, $E_{\rm avg}$, and n are plotted.

(from 6.9 to 13.1) at solar minimum and from 4.3 to 11.0 (from 4.0 to 9.2) at solar maximum (also see Figure 4). In solar cycle 24, the strahl (halo) κ ranges from 8.2 to 16.6 (from 7.2 to 14.0) at solar minimum and from 4.6 to 13.3 (from 4.3 to 11.1) at solar maximum. These results suggest that the acceleration of the solar wind suprathermal electrons is more efficient at solar maximum and in solar cycle 23 (also supported by the two-sample Kolmogorov–Smirnov test for Figure 4). In addition, the halo κ is generally smaller than the strahl κ, except during the solar minimum of cycle 23, probably implying a further acceleration for the halo.

For the 0.1–1.5 keV strahl and halo electrons at quiet times, the number density, n, is estimated by integrating the EESA electron measurements:

$$n = \sum_{0.1\ \mathrm{keV}}^{1.5\ \mathrm{keV}} \Omega \times \frac{J(E)}{v} \Delta E, \tag{2}$$

and the average electron energy, $E_{\rm avg}$ is defined as

$$E_{\rm avg} = \frac{\sum_{0.1\ \mathrm{keV}}^{1.5\ \mathrm{keV}} \Omega \times \frac{J(E) \times E}{v} \Delta E}{n}, \tag{3}$$

where the solid angle Ω is 4π for the isotropic halo and is $2\pi(1 - \cos 45^\circ)$ for the anisotropic strahl (that is generally present in two 22.5 PA bins at 1 au).

Figures 5 and 6 show the histograms of $E_{\rm avg}$ and n, respectively, for the 0.1–1.5 keV strahl and halo electrons in the selected quiet-time samples. The arrows indicate the median, with the horizontal lines (at the start of the arrow) showing the median absolute deviation. At quiet times, $E_{\rm avg}$ appears to be larger at solar minimum than at solar maximum in cycle 23, while $E_{\rm avg}$ is similar between solar minimum and maximum in cycle 24, for both strahl and halo electrons (also see Figure 8). These results are also supported by the two-sample Kolmogorov–Smirnov test (Press et al. 1992). The quiet-time strahl (halo) electrons at 0.1–1.5 keV contain ~0.02%–2% (~0.3%–5%) of the solar wind plasma density in solar cycle 23 and ~0.02%–0.7% (~0.2%–4%) of the solar wind plasma density in solar cycle 24. Both $E_{\rm avg}$ and n have no clear correlation with κ (not shown).

Figure 7 plots the temporal variation of κ, $E_{\rm avg}$, and n for the monthly average strahl (red) and halo (black) electrons at 0.1–1.5 keV during quiet times. At quiet times, κ shows a strong anti-correlation with the monthly sunspot number for

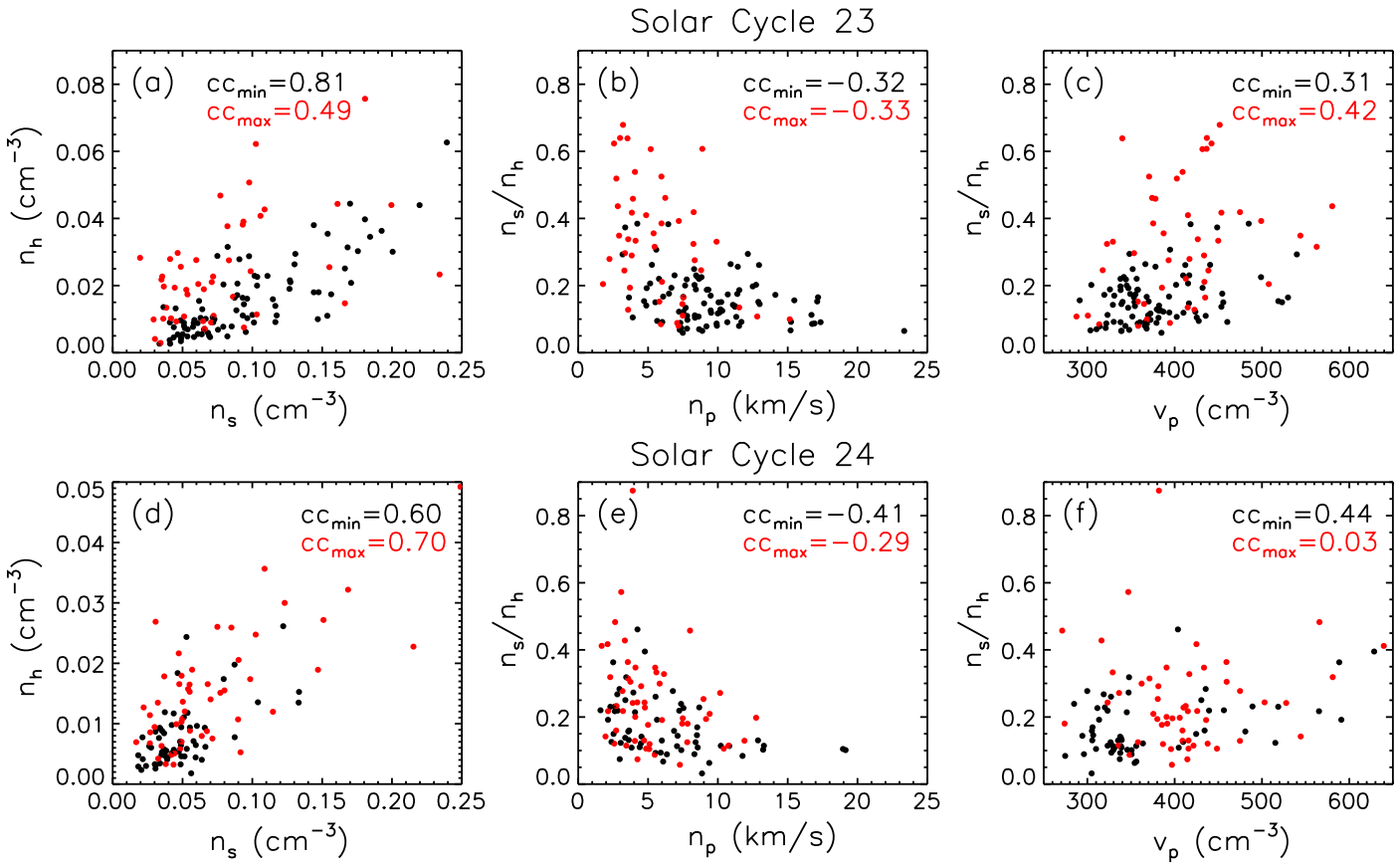


Figure 10. (a) and (d): scatter diagrams of the quiet-time halo density n_h vs. strahl density n_s observed at 0.1–1.5 keV at solar minimum (black) and maximum (red) in cycles 23 and 24. (b) and (c): scatter diagrams of the density ratio between n_s and n_h vs. n_p and v_p, for solar cycle 23. (e) and (f): same scatter diagrams, but for solar cycle 24. The estimated statistical uncertainties of n_s, n_h, and n_s/n_h are plotted (but smaller than the symbols).

both strahl and halo electrons. The number density n of strahl electrons has a noticeable correlation with the sunspot number, while the halo electron n shows no solar-cycle variation. In solar cycle 23 (see Figure 8), the quiet-time κ and $E_{\rm avg}$ appear to roughly correlate with the solar wind proton density n_p, but not correlate with the solar wind proton velocity v_p (both n_p and v_p can represent the characteristics of solar wind electron core population), for both strahl and halo electrons. The quiet-time n of strahl and halo electrons show no correlation with n_p and v_p. In solar cycle 24 (see Figure 9), the quiet-time κ, $E_{\rm avg}$, and n of strahl and halo electrons have no clear association with n_p and v_p.

Figure 10 shows that the number density ratio between the 0.1–1.5 keV strahl and halo electrons ranges from ∼6% to ∼38% (from ∼3% to ∼46%) at solar minimum and from ∼8% to ∼140% (∼6% to ∼87%) at solar maximum in cycle 23 (24). This ratio roughly anti-correlates with n_p, but correlates with v_p (except during the solar maximum of cycle 24). These results suggest that the strahl population has a different origin from the core population, while the generation of halo population is related to the strahl, probably depending on the properties of solar wind.

3. SUMMARY AND DISCUSSION

In this paper, we investigate the quiet-time energy spectrum of ∼0.1–1.5 keV strahl and halo electrons observed by the *WIND* 3DP instrument in the solar wind at the minimum and maximum of solar cycles 23 and 24. We find a strong positive correlation between the fitted κ and $T_{\rm eff}$ for both strahl and halo electrons, and a strong positive correlation between the number density n of strahl and halo electrons. For the ∼0.1–1.5 keV strahl electrons, κ (n) is mostly larger (smaller) at solar minimum than at solar maximum in both solar cycles. For the ∼0.1–1.5 keV halo electrons, κ also becomes larger at solar minimum, but n shows no difference between solar minimum and maximum. For both strahl and halo electrons, the average energy $E_{\rm avg}$ is generally larger at solar minimum in cycle 23, and is similar between solar minimum and maximum in cycle 24. Both the strahl n and halo n have no clear association with the solar wind core population, but the density ratio between strahl and halo electrons roughly anti-correlates (correlates) with the solar wind proton n_p (velocity v_p).

At 0.1–1.5 keV, the number density n of of strahl electrons shows a weak association with the sunspot number, while the n of halo electrons does not vary with the solar cycle (see Figure 7). On the other hand, both the strahl n and halo n have no association with the solar wind core population, while the halo n strongly correlates with the strahl n. These results support the idea that the strahl/halo have a different origin from the solar wind core: the strahl could originate from the Sun, e.g., due to the escaping electrons from the hot corona, while the halo may be due to some processes (e.g., scattering) acting on the strahl in the IPM (e.g., Montgomery et al. 1968; Feldman et al. 1975; Rosenbauer et al. 1977; Pilipp et al. 1987; Pierrard et al. 2001).

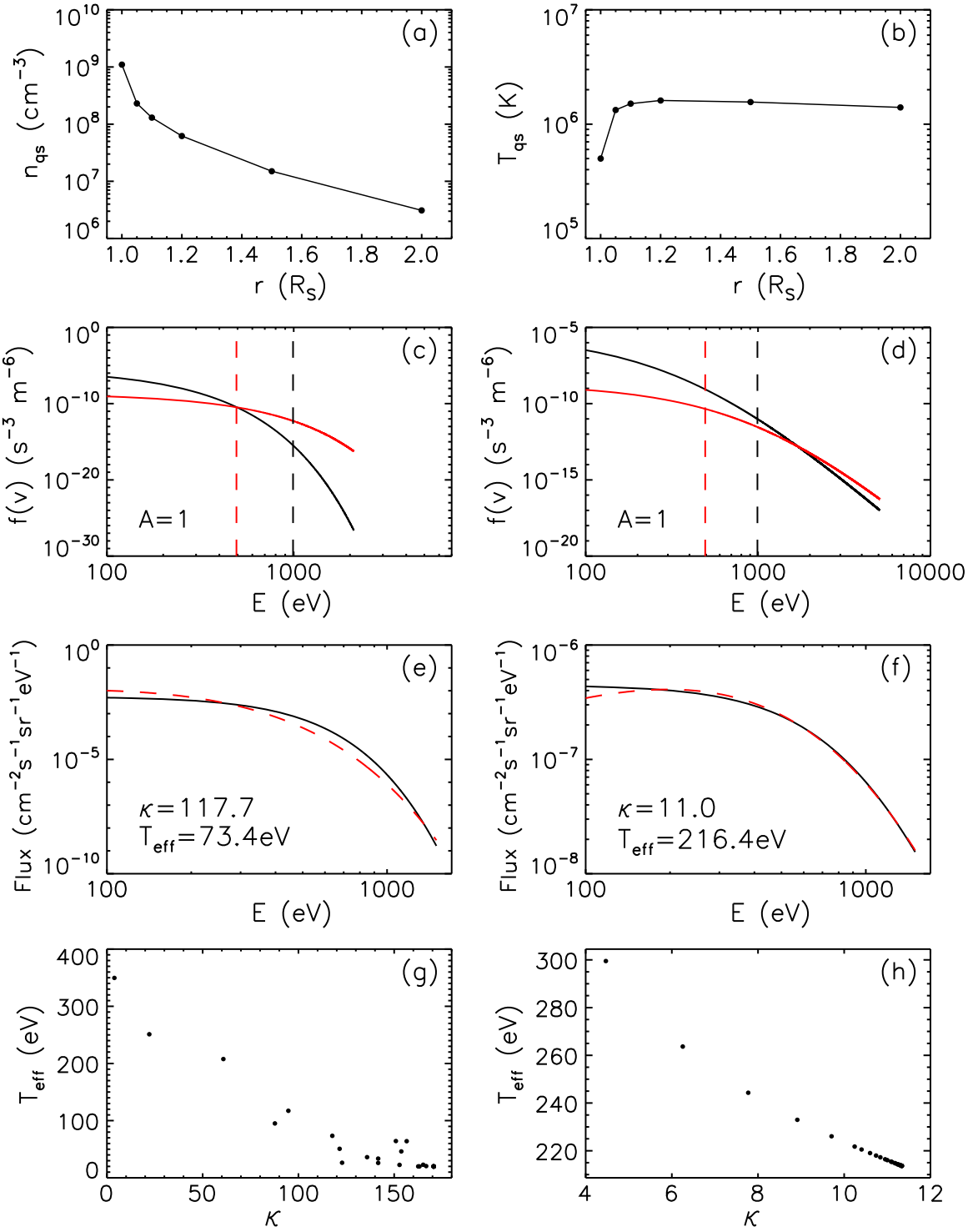


Figure 11. (a) and (b): the quiet-sun density and temperature from 1 R_S to 2 R_S. (c) and (d): the electron Maxwellian (left)/Kappa (right) velocity distribution at 1 R_S (black) and 2 R_S (red), for $A = 1$. (e) and (f): the simulated electron spectrum at 1 au, for the escaping electrons with a Maxwellian (left) or Kappa (right) distribution. The red dashed lines represent a Kappa function fit. (g) and (h): the simulated $T_{\rm eff}$ vs. κ at 1 au, for the escaping electrons with a Maxwellian (left) or Kappa (right) distribution.

For the 0.1–1.5 keV strahl electrons at quiet times, κ ranges from 4.3 to 15.3 (from 4.6 to 16.6) in solar cycle 23 (24), anti-correlated with the sunspot number (see Figure 7). This suggests a more efficient acceleration for the ~0.5–1.5 keV electrons occurring at solar maximum and in cycle 23. In solar cycle 23, κ roughly correlates with the solar wind proton density n_p, but not with the solar wind proton velocity v_p. In solar cycle 24, κ has no obvious assoication with both n_p and v_p. These results imply that the generation of strahl electrons can differ in different solar cycles.

The positive correlation between κ and $T_{\rm eff}$ is likely formed in the generation process of strahl electrons at the Sun. However, our simple model of the escaping electrons from the corona cannot produce such a positive correlation. Here we

assume that the velocity distribution function of electrons escaping from a source at r (the heliocentric distance) is a Maxwellian distribution or a Kappa distribution (with a given $\kappa = 9$), determined by the density $n(r) = n_{qs}(r)$ and $T(r) = A \times T_{qs}(r)$, where n_{qs} and T_{qs} are the quiet-Sun density and temperature defined by Withbroe (1988; see Figures 11(a) and (b)). During their propagation, the escaping electrons will lose energy to Coulomb collisions and to the ambipolar electrostatic potential (Wang et al. 2006): $dE/dr = (dE/dr)_{\rm col} + (dE/dr)_{\rm AEP} = -1.82 \times 10^{-7} n(r)/E - 0.994/r^2$, where E is the electron energy in keV and r is the heliocentric distance in R_S. After considering the electron escaping from 1 R_S to 2 R_S, we estimate the spectrum of electrons arriving at 1 au as

$$f(E_{1\ \rm au}) = \int_{1.0R_S}^{2.0R_S} \frac{f(E(r))}{2.0R_S - 1.0R_S} \left(\frac{r}{1\ \rm au}\right)^2 dr \qquad (4)$$

for a temperature coefficient A varying from 0.8 to 1.3, and fit the modeled electron spectrum at 1 au to a Kappa function (see Figures 11(e) and (f)). We find that the escape of electrons from different coronal regions with different A shows a negative correlation between κ and $T_{\rm eff}$ (see Figures 11(g) and (h)), in contradiction to the observed positive correlation (Figure 3). Further investigations will require a very careful and detailed modeling including other effects, e.g., the exospheric models (e.g., Pierrard et al. 2001; Pierrard & Pieters 2014).

For the quiet-time halo electrons, κ ranges from 4.0 to 14.0, with smaller values occurring at solar maximum and in cycle 23. In the selected samples, the halo κ is generally smaller than the strahl κ (except during the solar minimum of cycle 23), indicating the presence of further acceleration, as well as the PA scattering, in the generation of halo. In solar cycle 23, the halo κ index is roughly correlated with the proton density n_p, but not correlated with the proton velocity v_p. In solar cycle 24, the halo κ index has no correlation with n_p and v_p.

In both solar cycles, the density ratio between the quiet-time strahl and halo electrons roughly anti-correlates (correlates) with n_p (v_p, except during the solar maximum of cycle 24), consistent with previous observations that the strahl is predominantly observed in fast solar wind (e.g., Pilipp et al. 1987). This suggests that the generation of halo electrons could be related to the properties of solar wind, as well as its propagation, in the IPM. Various theories proposed that the halo is formed by strahl electrons interacting with waves such as Langmuir waves and whistler waves (e.g., Vocks et al. 2005; Yoon et al. 2006; Kim et al. 2015). We will investigate the comprehensive relation between the halo and the properties of solar wind plasma and turbulence in the future.

This research at Peking University is supported in part by the NSFC under contracts 41421003, 41274172, 41474148, 41231069, 41574168, and 41474147. R.W.S. thanks the German space agency DLR for support under grants 50 OT 1202 and 50 OC 1501 and also thanks ISSI for its hospitality and support of parts of this work. The work at UC Berkeley is supported by NASA grants NNX13AL11G and NNX14AC07G. The work at UAH is supported by NASA grants NNX14AC08G and NNX15AJ93G.

REFERENCES

Che, H., & Goldstein, M. L. 2014, ApJL, 795, L38
Feldman, W. C., Asbridge, J. R., Bame, S. J., Montgomery, M. D., & Gary, S. P. 1975, JGR, 80, 4181
Hammond, C. M., Feldman, W. C., McComas, D. J., Phillips, J. L., & Forsyth, R. J. 1996, A&A, 316, 350
Kim, S., Yoon, P. H., Choe, G. S., & Wang, L. 2015, ApJ, 806, 32
Lepping, Acũna, M. H., Burlaga, L. F., et al. 1995, SSRv, 71, 207
Lin, R. P. 1998, SSRv, 16, 189
Lin, R. P., Anderson, K. A., Ashford, S., et al. 1995, SSRv, 71, 125
Livadiotis, G., & McComas, D. J. 2010, ApJ, 714, 791
Maksimovic, M., Pierrard, V., & Lemaire, J. F. 1997a, A&A, 324, 725
Maksimovic, M., Pierrard, V., & Riley, P. 1997b, GeoRL, 24, 1151
Maksimovic, M., Zouganelis, I., Chaufray, J.-Y., et al. 2005, JGR, 110, A09104
Montgomery, M. D., Bame, S. D., & Hundhausen, A. J. 1968, JGR, 73, 4999
Pierrard, V., Maksimovic, M., & Lemaire, J. 1999, JGR, 104, 17021
Pierrard, V., Maksimovic, M., & Lemaire, J. 2001, Ap&SS, 277, 195
Pierrard, V., & Pieters, M. 2014, JGRA, 119, 9441
Pilipp, W. G., Miggennieder, H., Montgomery, M. S., et al. 1987, JGR, 92, 1075
Press, W. H., Teukolsky, S. A., Vetterling, W. T., & Flannery, B. P. 1992, Numerical Recipes in C. The Art of Computing (Second ed.; Cambridge, UK: Cambridge Univ. Press)
Rosenbauer, H., Schwenn, R., Marsch, E., et al. 1977, JGZG, 42, 561
Salem, C., Bale, S. D., & Maksimovic, M. 2007, in Proc. Second Solar Orbiter Workshop (Athens, Greece: ESA Publications), S-641
Salem, C., Bosqued, J.-M., Larson, D. E., et al. 2001, JGR4, 106, 21701
Scudder, J. D. 1992, ApJ, 398, 299
Scudder, J. D., & Olbert, S. 1979, JGR, 84, 6603
Stverak, S., Maksimovic, M., Travnicek, P. M., et al. 2009, JGR, 114, A05104
Vasyliunas, V. M. 1968, JGR, 73, 2839
Vocks, C., Salem, C., Lin, R. P., & Mann, G. 2005, ApJ, 627, 540
Wang, L. 2009, PhD thesis, Univ. California
Wang, L., Lin, R. P., Krucker, S., & Gosling, J. T. 2006, GeoRL, 33, L03106
Wang, L., Lin, R. P., Salem, C., et al. 2012, ApJL, 753, L23
Wang, L., Yang, L., He, J., et al. 2015, ApJL, 803, L2
Withbroe, G. L. 1988, ApJ, 325, 442
Yang, L., Wang, L., Li, G., et al. 2015, ApJL, 811, L8
Yoon, P. H., Rhee, T., & Ryu, C.-M. 2006, JGR, 111, A09106
Zouganelis, I. 2008, JGR, 113, A08111
Zouganelis, I., Maksimovic, M., Meyer-Vernet, N., Lamy, H., & Issautier, K. 2004, ApJ, 606, 542